# Econ21003-Studyguide

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October 14, 2026

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# Introduction

The begining is the most important part of a journey!

- Syllabus
- Course Calendar
- Ten Principles of Economics

Continue to the next page to see exam 1 study guide page!

# 1 *Symbols, abbr. and formula*

***Symbols and abbr***

- $L$ = Labor units, in persons or in hours
- $K$ = Amount of physical capital
- $H$ = Amount of human capital
- $N$ = Amount of Natural resources
- $MB$ = Marginal Benefit; $MC$ = Marginal Cost; $MP$ = Marginal Product
- $HH_s$ *or* $hh_s$ = house holds
- $PPF$ = Production possibility Frontier
- $D$ = Demand curve
- $GDP$ = Gross Domestic Product
- $GNP$ = Gross National Product
- $Y$ = Consumers' income, National Income, or GDP
- $C$ = consumption; I = investment; $G$ = Government purchases;
- $NX$ = Net exports, which is (X-M) where $X$ = exports; $M$ = imports
- $CPI$ = Consumer Price Index
- $Y/L$ = Output per person, (Labor) Productivity
- $K/L$ = Capital per worker (per person)
- $H/L$ = Human capital per person
- $i$ = Nominal interest rate; $r$ = Real interest rate; $\pi$ Inflation or sometimes profit
- $P$ = Price of a good or the price level
- $G$ = government purchases ; $G\&S$ = Goods and services ; $Govt$ = government
- $Q$ = Quantity or amount of the good or service
- $S_{pvt}$ = Private saving
- $S_{gov}$ *or* $S_{pub}$ = Public or government saving
- $T$ = Net taxes, which is (Taxes-govt transfers)
- $RGDP$ = Real GDP
- $NGDP$ = Nominal GDP
- $S$ = Supply Curve, or National Saving (in another context)
- $W$ = Nominal Wage
- $w$ = Real wage
- $A$ = Technology or Total factor productivity
- $TFP$ = Total factor productivity

***Formula****

- growth of variable X $= (\frac{X_{yr2} - X_{yr1}}{X_{yr1}}) \cdot 100 = (\frac{X_{yr2}}{X_{yr1}} - 1) \cdot 100$
- $Y = C + I + G + NX = C + I + G + X - M$
- $NGDP =$ Output of a given year valued at that year's prices
- $RGDP =$ Output of a given year valued at base year prices
- $GDP\ deflator = \frac{NGDP}{RGDP} \times 100$
- $$\begin{aligned} GNP &= GDP + \text{Net factor payments from abroad} \\ &= GDP + \text{Income of domestic factors earned abroad} \\ &\quad - \text{Income of foreign factors earned at home} \end{aligned}$$
- $CPI = \frac{\text{Cost of basket using current prices}}{\text{Cost of basket using base year prices}} \times 100$
- Number of years for a variable to double $\approx \frac{70}{\text{Growth rate of the varible}}$
- Amount in Today's dollar = Amount in year T $\times \frac{\text{Price level today}}{\text{Price level in year T}}$
- $r = i - \pi$
- $Y_d = Y - TR - T =$ disposable income

For closed economy:

- $S_{pvt} = Y + TR - T - C$
- $S_{gov} = T - TR - G$
- $S = S_{pvt} + S_{gov} = Y - C - G$
- $S = I$
- $u - rate = 100 \times \frac{\#Unemployed}{\#Labor\ Force}$
- $L.F\ partic.\ rate = 100 \times \frac{\#Labor\ Force}{\#Adult\ popn}$

# 2 Review for Exam 1

## 2.1 Key Concepts and Definitions

***Chapter 1***

- *Three* key economic ideas
- Efficiency vs Equality
- Technical vs Allocative efficiency
- Positive vs Normative Analysis
- Types of Economies
- Role of economic models
- Micro vs. Macro

***Chapter 2***

- Production Possibilities Frontier (PPF)
- Opportunity Cost
- Comparative advantage vs Absolute advantage
- Complete specialization
- Gains from trade
- Market system
- Circular flow diagram

***Chapter 3***

- Perfectly competitive markets
- Law of Demand / Law of Supply
- Supply and Demand curves vs. quantity supplied and quantity demanded
- Supply and Demand schedules
- Effect of change in price – change in quantity supplied/demanded
- Factors that shift Demand (external factors of Demand)
- Factors that shift Supply (external factors of Supply)
- Substitute vs complement and normal vs inferior
- Equilibrium – identifying and interpreting
- Interpreting shifts and double shifts

## 2.2 Sample Multiple Choice questions

## 2.3 Worksheet

1. Consider the following scenario: Saudi Arabia and the United States each have 100 worker hours per week to produce oil, corn, or a combination of both. The following table shows their production possibilities:

| Country | Oil using 100 hours (barrels) | Corn using 100 hours (bushels) |
|---|---|---|
| Saudi Arabia | 100 | 25 |
| United States | 50 | 100 |

   a. What is Saudi Arabia's opportunity cost of producing: 1 barrel of oil? _____ 1 bushel of corn? _____
   b. What is the United States' opportunity cost of producing: 1 barrel of oil? _____ 1 bushel of corn? _____
   c. Who has the absolute advantage in oil production? _____
   d. Who has the absolute advantage in corn production? _____
   e. Who has the comparative advantage in oil production? _____
   f. Who has the comparative advantage in corn production? _____

Answer Key

a. 1/4; 4
b. 2; 1/2
c. Saudi Arabia
d. USA
e. Saudi Arabia
f. USA

2. Use the information from Question 1. Graph the Production Possibilities Frontier for both countries.

Answer Key

With oil on the x-axis and corn on the y-axis, Saudi Arabia's PPF has an x-intercept of 100 and a y-intercept of 25. The United States' PPF has an x-intercept of 50 and a y-intercept of 100. Both PPFs are linear, assuming that the opportunity cost is constant at every point along each PPF.

3.

a. Fill in the following table assuming *complete specialization* and a trade action of: Saudi Arabia trades 45 barrels of oil for 40 bushels of corn.

| | **Saudi Arabia** | | **United States** | |
|---|---|---|---|---|
| | Oil (barrels) | Corn (bushels) | Oil | Corn |
| **Without Trade** | | | | |
| Production | 50 | 12.5 | 25 | 50 |
| Consumption | 50 | 12.5 | 25 | 50 |
| **With Trade** | | | | |
| Production | | | | |
| Trade Action | | | | |
| Consumption | | | | |
| **Gains from Trade** | | | | |
| Increase in Consumption | | | | |

Answer Key

b. The following table shows production and consumption of two countries for Lemonade and Pizza. If the terms of trade is 193 Pitchers of Lemonade to 110 pizzas, fill in the blank given by the letters if each country completely specializes in the good in which it has the applicable advantage.

| | U.S. | | China | |
|---|---|---|---|---|
| | Lemonade in Pitchers | Pizza | Lemonade in Pitchers | |
| Production & Consumption without Trade | 200 | 100 | 180 | |
| Production with trade | 400 | 0 | 0 | |
| Trade Action | a | b | c | |
| Consumption with trade | e | f | g | |
| Gains from trade | i | j | k | |

Answer Key

4. Graph the supply and demand curves using the following supply and demand schedule. Label the equilibrium price and quantity.

| Price | Quantity Demanded | Quantity Supplied |
|---|---|---|
| $4 | 10,000 tickets | 6,000 tickets |
| 8 | 9,000 | 7,000 |
| 12 | 8,000 | 8,000 |
| 16 | 7,000 | 9,000 |
| 20 | 6,000 | 10,000 |

- Answer if there is shortage or surplus at each price of the good

Answer Key

| Price | Qd | Qs | Shortage |
|---|---|---|---|
| 4 | 10000 | 6000 | 4000 |
| 8 | 9000 | 7000 | 2000 |
| 12 | 8000 | 8000 | 0 |
| 16 | 7000 | 9000 | -2000 |
| 20 | 6000 | 10000 | -4000 |

5. Effect of Shift of Demand and Supply on Eq'm Price and Quantity:

Each of the events listed below has an impact on the market for smartphones. For each event, identify which curve is affected (supply or demand or both for smartphones), the direction in which it shifts, and the resulting impact on the equilibrium price and quantity of smartphones.

a. The price of tablets increases.
b. Consumers' incomes decrease, and smartphones are a normal good.
c. The price of semiconductor chips used to manufacture smartphones increases.
d. A new social trend increases consumers' preference for smartphones.
e. Consumers expect the price of smartphones to fall in the future.
f. A technological advance in the manufacture of smartphones occurs.
g. The prices of smartphone cases and wireless earbuds are reduced.

Answer Key

a. Demand shifts right, equilibrium price goes up, quantity goes up.

b. Demand shifts left, equilibrium price goes down, quantity goes down.

c. Supply shifts left, equilibrium price goes up, quantity goes down.

d. Demand shifts right, equilibrium price goes up, quantity goes up.

e. Demand shifts left, equilibrium price goes down, quantity goes down (currently).

f. Supply shifts right, equilibrium price goes down, quantity goes up.

g. Demand shifts right, equilibrium price goes up, quantity goes up.

# 3 Review for Exam 2

## 3.1 Key Concepts and Definitions

***Chapter 8: GDP***

- Gross Domestic Product (GDP) — what does it measure?
- Final good and intermediate good
- Total Production = Total Income = Total Spending
- Components of GDP: What is included in C, I, G, NX?
- Limitations or imperfections of GDP
- Calculating Nominal GDP, Real GDP, and the GDP Deflator
- Calculating inflation from Nominal and Real GDP
- Other measures of total production and total income: GNP, National income, disposable income, etc

***Chapter 9: Unemployment and Inflation***

- How we segment the population with regards to Unemployment
- Household survey and employment survey
- Calculating unemployment rate, labor force participation rate, and the employment-population ratio
- Discouraged workers
- Distinguishing the three types of unemployment: Frictional, Structural and cyclical
- Full employment and the Natural Rate of Unemployment
- Calculating the Consumer Price Index (CPI)
- Calculating inflation
- Indexation
- Shortcomings or imperfections of CPI: Substitution Bias, Quality Bias, New goods Bias and Outlet Bias
- Nominal vs. Real Interest rates and their relationship
- Uses of CPI: Converting prices to today's dollars, converting nominal variables to real variables

***Chapter 10: Longrun Economic Growth, Saving + Investment, Business Cycle***

- Distingish GDP, real GDP and real GDP per capita
- Calculating growth from one year to the next
- "Constant growth rate rule": given the economy grows at different rate for several years, what would be the equivalent constant rate of growth?
- Two ways of average growth rates: simple and compound
- Rule of 70
- Determinants of long-run economic growth
- Actual vs. Potential GDP
- Financial System: financial markets vs. financial intermediaries
- Direct finance vs. indirect finance
- Savings (private, government and national/aggregate saving) and Investment Identity
- Distinguish the difference between Saving and investment
- The Market for Loanable Funds-– Analyzing effects of shifts and movements along.
- Crowding out

## 3.2 Sample Multiple Choice Questions

## 3.3 Workout Practice (Interactive)

# A Introduction to Personal Finance and Financial Independence

I can calculate the motion of heavenly bodies, but not the madness of people.

– Often attributed to Sir Isaac Newton, reflecting on his financial losses during the South Sea Bubble of 1720.

Suggested Readings:

- Collins, J. L. (2016). The Simple Path to Wealth: Your Road Map to Financial Independence and a Rich, Free Life.

Intermediate Reading on Investing:

- Malkiel, B. G. (2019). A Random Walk Down Wall Street: The Best Investment Guide That Money Can Buy (13th ed.). W. W. Norton & Company.
- Video:

The Simple Path to Wealth JL Collins, Talks at Google

- Blog for general economics from my favorite economist:

Greg Mankiw Blog

- Personal finance is about more than earning a good salary. It is about how you manage the money you earn and the decisions you make today that affect your future. Two people can earn the same income but end up in very different financial situations because they make different choices about spending, debt, and saving.

My goal is for you to understand a few basic principles that can help you become financially independent without sweating it and avoid unnecessary financial stress, and build wealth over time.

## A.1 Live Within Your Means.

The first rule of personal finance is simple: Spend less than you earn.

It may sound obvious, but many people struggle financially because they spend more than their income allows. They purchase things they cannot afford, rely on credit cards, or borrow money to maintain a lifestyle beyond their means.

Suppose you earn $2,000 per month after taxes. You need to pay rent, buy food, cover transportation costs, pay your phone bill, and meet other expenses. If you spend the entire $2,000 every month, you have nothing left to save. If an unexpected expense occurs, you may have to borrow money.

A better approach is to create a budget. A budget is simply a plan for how you will use your income.

For example, you might allocate your monthly income to housing, food, transportation, entertainment, savings, and debt repayment. The amounts will depend on your circumstances, but you should deliberately set aside some money for savings rather than waiting to see what remains at the end of the month.

Economics teaches us about opportunity cost: when you spend money on one thing, you give up the opportunity to spend it on something else.

Every time you buy something, ask yourself: Do I really need this? What else could I do with this money? Will this purchase help me achieve my long-term goals?

You do not have to avoid every enjoyable purchase. The objective is to spend intentionally rather than impulsively. Living within your means gives you greater freedom, reduces financial stress, and helps you prepare for the future.

## A.2 Save for Emergencies and Future Goals.

Saving is an essential part of financial security. You should not depend entirely on your next paycheck to handle unexpected expenses.

Imagine that your car needs a $700 repair, or you suddenly lose your job. If you have no savings, you may have to use a credit card or borrow money. If you have an emergency fund, you can handle the situation without immediately going into debt.

Start by saving a manageable amount regularly, even if you can afford only $20 or $50 per paycheck. As your income increases, try to increase your savings as well.

Over time, work toward building an emergency fund that can cover approximately three to six months of essential living expenses. The appropriate amount depends on your circumstances, job security, and responsibilities.

Keep emergency money somewhere safe and accessible, such as an interest-bearing savings account. Do not put money you may need for an emergency into stocks, because stock prices can fall when you need to withdraw the money.

It is also helpful to save for specific goals, such as buying a car, making a down payment on a house, pursuing further education, or preparing for retirement.

Remember that saving is not simply about having money left over. It is about making a deliberate choice to protect your future.

## A.3 Understand the Power of Compound Interest.

One of the most powerful ideas in personal finance is compound interest.

Compound interest occurs when you earn returns on your original savings and then earn additional returns on the accumulated growth. Over time, this process can significantly increase the value of your money.

Consider a simple example. Suppose you invest $100 every month and earn an illustrative average annual return of 7%, compounded monthly. If you continue investing for 30 years, you would contribute a total of $36,000. Your account could grow to approximately $122,000 before taxes, fees, and inflation, assuming that constant rate of return.

The actual returns from investments are not guaranteed, and the value of your investments can decline. Nevertheless, this example illustrates the potential power of long-term investing.

Notice that you do not need to begin with thousands of dollars. Regular contributions, time, and patience can make a significant difference.

This is why starting early can be so valuable. If you begin saving and investing while you are young, your money has more time to grow. Waiting until later in life means you may need to contribute more to reach the same financial goal.

You cannot control the market's future returns, but you can control how regularly you save, how much you contribute, and how carefully you manage investment costs.

## A.4 Learn How to Invest in Stocks and Bonds.

Saving helps protect your money, but "investing" can help you build wealth over the long term. Two important types of investments are stocks and bonds.

Stocks, which is a partial ownership of a company has higher average return but also higher risk because its price flactuates. For many beginners, a diversified, low-cost stock index fund is

worth investigating. Rather than investing all your money in one company, an "index fund" can give you exposure to many companies at once.

Diversification reduces the impact of problems at an individual company, although it cannot eliminate the risk of a broad market decline. Historically, a diversified stock index such as the S&P 500 has returned an average of about 10-12% per year, which means that your savings double every 7 years.

Bonds: When you purchase a bond, you are generally lending money to a government or company. In return, the issuer promises to make payments according to the bond's terms and typically repay the principal at maturity.

A well ***diversified Bonds*** such as the "Fidelity U.S. Bond Index Fund" have lower average return (3% - 6%) but are less volatile/risky. So, bonds can provide income and help diversify a portfolio than stocks.

The rule of thumb is that you have to have 30-40% of your saving in bonds and 60-70% in stock even though this allocation can vary with age. The nearer to retirement the more bonds you should have.

***Rebalancing***: Suppose you are keeping 40% bond and 60% stock portfolio and stock prices decrease by 50% as in 2008, you allocation automatically becomes 70% bond and 30% stock. Then rebalancing to 40% bond and 60% stock makes you richer when the price of stock rebounds as in 2009 and beyond.

If you are investing in the United States, you can investigate investment accounts such as a regular brokerage account or an eligible retirement account, such as a Roth IRA or workplace 401(k). Understand the account's fees, tax rules, investment choices, and withdrawal restrictions before investing.

A sensible starting point is to establish emergency savings, address expensive debt, and then consider regular investments suited to your goals. Money you need in the next year or two generally should not depend on the stock market performing well.

## A.5 Avoid High-Interest Debt and Financial Speculation.

Debt can help people pay for important purchases, education, or a home. However, borrowing becomes dangerous when interest costs grow faster than a person's ability to repay the loan.

Suppose you owe $2,000 on a credit card charging 20% annual interest. If you make no payments and receive no additional charges, the balance can double every 3.5 years.

The same principle of compounding that helps investments grow can work against you when you owe money. Paying only the minimum on a credit card can keep you in debt for a long time.

You should also be cautious about speculative investments. An investment is not automatically a good opportunity just because its price is rising. If you buy something only because you expect someone else to pay more for it, you may be relying on speculation rather than a sound investment decision.

Consider the experience of Sir Isaac Newton during the South Sea Bubble of 1720. Newton initially made money from his investments in the South Sea Company, but he later bought back into the market as prices rose. When the bubble collapsed, he suffered substantial financial losses.

You might be required to put your savings in investment banks like Fidelty or TIAA, but you should manage them yourself. Never give your saving to "money managers"– remember the ponzi schem Travis Kelce and others fell into recently: https://www.reuters.com/sports/travis-kelce-named-victim-35m-ponzi-scheme–flm-2026-09-16/ and also Bernie Madoff, the financier behind one of the largest Ponzi schemes in history.

Even a brilliant person can be influenced by excitement, overconfidence, and the fear of missing out.

Before investing, ask yourself whether you understand what you are buying, what risks you are taking, and how the investment fits your financial goals and how it is managed. Never invest money you cannot afford to lose in a speculative venture, and be especially cautious about borrowing to invest.

Conclusion:

- Build Good Financial Habits Early Your financial future will be shaped by many decisions you make over time. You do not need to become wealthy overnight, predict every movement in the stock market, or earn a huge salary to begin managing your money effectively.
- Start with a few basic habits:
  - Spend less than you earn.
  - Set aside money regularly for emergencies and future goals.
  - Start investing early when your financial circumstances allow.
  - Understand the differences between stocks and bonds.
  - Diversify your investments rather than betting everything on one company.
  - Avoid unnecessary high-interest debt.
  - Make financial decisions based on your goals rather than fear, excitement, or pressure from others.

The most important lesson is that financial security is built through consistent decisions, not one lucky investment.

As you study economics, remember that personal finance is an everyday application of the subject. Scarcity forces you to make choices, opportunity cost helps you evaluate alternatives,

compound interest shows the value of time, and economic incentives influence your spending and saving behavior.

## A.6 Important Formula

***Symbols, abbr. and formula*** Symbols and abbr. $L$ = Labor units, in persons or in hours $K$ = Amount of physical capital $H$ = Amount of human capital $N$ = Amount of Natural resources $MB$ = Marginal Benefit; $MC$ = Marginal Cost; $MP$ = Marginal Product $HH_s$ *or* $hh_s$ = house holds $PPF$ = Production possibility Frontier $D$ = Demand curve $GDP$ = Gross Domestic Product $GNP$ = Gross National Product $Y$ = Consumers' income, National Income, or GDP $C$ = consumption; I = investment; $G$ = Government purchases; $NX$ = Net exports, which is (X-M) \ where $X$ = exports; $M$ = imports $CPI$ = Consumer Price Index $Y/L$ = Output per person, (Labor) Productivity $K/L$ = Capital per worker (per person) $H/L$ = Human capital per person $i$ = Nominal interest rate; $r$ = Real interest rate; $\pi$ Inflation or sometimes profit $P$ = Price of a good or the price level $G$ = government purchases ; $G\&S$ = Goods and services ; $Govt$ = government $Q$ = Quantity or amount of the good or service $S_{pvt}$ = Private saving $S_{gov}$ *or* $S_{pub}$ = Public or government saving $T$ = Net taxes, which is (Taxes-govt transfers) $RGDP$ = Real GDP $NGDP$ = Nominal GDP $S$ = Supply Curve, or National Saving (in another context) $W$ = Nominal Wage $w$ = Real wage $A$ = Technology or Total factor productivity $TFP$ = Total factor productivity

***Formula****
$Slope = \frac{Rise}{Run} = \frac{\Delta Y}{\Delta X}$ $Opportunity\ cost\ of X = |\frac{Amount\ of\ Y\ given\ up}{Amount\ of\ X\ obtained}| = |\frac{\Delta Y}{\Delta X}|$ Standard method of percentage Change:\
$\frac{End\ Value - Start\ Value}{Start\ Value} \times 100$ \ e.g., growth of variable X $= 100 \cdot (\frac{X_2 - X_1}{X_1}) = 100 \cdot (\frac{X_2}{X_1} - 1)$
$Y = C + I + G + NX = C + I + G + X - M$ $GDP = sum\ of\ P \times Q$, for all goods $GDP\ deflator = \frac{NGDP}{RGDP} \times 100$ $GNP = GDP$ + Net factor payment from abraod \ = $GDP$ + Income of domestic factors earned abroad \ −income of Foreign factors earned at home
%if $Z = X \cdot Y \Rightarrow \%\Delta Z \approx \%\Delta X + \%\Delta Y$ %if $Z = X/Y \Rightarrow \%\Delta Z \approx \%\Delta X - \%\Delta Y$ \ %For example, since $Total\ Revenue = P \times Q \Rightarrow \%\Delta TR = \%\Delta P + \%\Delta Q$ After

chapter 13: $u - rate = 100 \times \frac{\#Unemployed}{\#Labor\ Force}$ $L.F\ partic.\ rate = 100 \times \frac{\#Labor\ Force}{\#Adult\ popn}$ $Cyclical\ unemployment = u - u^*$ where, $u$ =actual unempl and $u^*$ = natural or full employment unemployment $Value\ of\ money = 1/P$ $M \cdot V = P \cdot Y$ $Aggr.\ demand: Y_d = C + I + G + NX$ $Aggr.\ supply: Y_S = Y_N + a(P - P^e)$ $CPI = \frac{\text{Cost of basket using current prices}}{\text{Cost of basket using base year prices}} \times 100$ Number of years a variable to double $\approx \frac{70}{\text{Growth rate of the varible}}$ $Amount\ in\ Today's\ dollar = Amount\ in\ year\ T \times \frac{\text{Price level today}}{\text{Price level in year T}}$ $r = i - \pi$ and $r^e = i - \pi^e$ $real\ wage = W/P$ $Prod.\ function: Y = AF(L, K, H, N)$ $Y_d = Y - T =$ disposable income For closed economy: \ $S_{pvt} = Y - T - C$ \ $S_{gov} = T - G$ \ $S = S_{pvt} + S_{gov} = Y - C - G$\

# B

Coming soon